

Comprehensive Road Scene Understanding for Autonomous Driving: From Mask-Architecture Segmentation to Anomaly Detection

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Abstract

Reliable perception in open-world driving requires both accurate dense segmentation of known categories and the ability to flag unknown, out-of-distribution (OoD) objects that were never seen during training. This report studies road-scene understanding through the lens of modern mask-classification architectures, taking the Encoder-only Mask Transformer (EoMT) with a DINOv2 backbone as the central model. We first review the progression from real-time per-pixel networks (ERFNet) and the panoptic formulation to MaskFormer, Mask2Former and EoMT. We then (i) compare two provided EoMT checkpoints—one trained on Cityscapes for semantic segmentation and one on COCO for panoptic segmentation—under a single, fair 16-class semantic protocol obtained by projecting COCO predictions into the Cityscapes label space; (ii) fine-tune the COCO model on Cityscapes with a staged freezing schedule, recovering a validation mIoU of 67.3% with the backbone frozen; and (iii) evaluate post-hoc anomaly segmentation with Maximum Softmax Probability, Max-Logit, Max-Entropy and, for the mask architecture, Rejected-by-All (RbA), across four public benchmarks. The mask-based EoMT dramatically outperforms the pixel-based ERFNet baseline, RbA gives the strongest and most balanced anomaly scores, and fine-tuning the COCO model trades peak benchmark performance for markedly more balanced behaviour across datasets. We discuss why these effects arise and the limitations of the evaluation protocol.

1. Introduction

1.1. Project context

Scene understanding for autonomous driving is commonly cast as a dense labelling problem. Semantic segmentation assigns a class label to every pixel; instance

segmentation additionally separates individual objects of the same class; and panoptic segmentation [10] unifies the two by labelling every pixel with both a class and, for countable “thing” classes, an instance identity. Deep networks have made these tasks remarkably accurate in structured, closed-world conditions where the set of categories is fixed in advance.

Real driving, however, is an open world. Networks trained on a closed label set are by construction unable to represent objects outside that set, yet such unknown, anomalous or out-of-distribution (OoD) objects—a lost cargo box, an animal, debris on the road—are exactly the cases where a perception failure is most dangerous [1, 2]. Detecting and localising these regions is therefore a safety-critical complement to ordinary segmentation. This project follows a single thread through both problems: it studies the dominant mask-classification family of segmentation models, compares two pre-trained variants of a state-of-the-art member of that family, adapts one of them to the driving domain by fine-tuning, and finally repurposes the same models for anomaly segmentation using training-free (post-hoc) scoring rules.

1.2. Literature foundations

From per-pixel to mask classification. Efficient per-pixel networks such as ERFNet [15] established that accurate real-time semantic segmentation is feasible on embedded automotive hardware by factorising convolutions for a favourable accuracy/latency trade-off. Such models treat segmentation as independent per-pixel classification. MaskFormer [3] challenged this view: it shows that predicting a set of binary masks, each paired with a single class label, is general enough to solve semantic, instance and panoptic segmentation with one model, loss and training procedure. Mask2Former [4] refined this idea with masked attention, restricting cross-attention to the region of each predicted mask, which improves both accuracy and convergence and

made a single architecture competitive across all three tasks.

Foundation backbones and EoMT. In parallel, self-supervised pre-training produced strong general-purpose visual features: DINOv2 [14] learns robust representations from unlabelled images that transfer across many dense tasks. The Encoder-only Mask Transformer (EoMT) [9] brings these threads together: it argues that a sufficiently large, well pre-trained plain Vision Transformer does not need the task-specific adapters, pixel decoders and Transformer decoders of Mask2Former. Instead, learnable queries are processed inside the ViT encoder itself, yielding a simpler architecture that matches specialised models while being substantially faster. EoMT is the central model of this project, instantiated with a DINOv2 ViT backbone.

Anomaly segmentation and post-hoc scoring. The anomaly-segmentation task and its evaluation are defined by the SegmentMeIfYouCan (SMIYC) [2] and Fishyscapes [1] benchmarks, together with the earlier RoadAnomaly dataset [12]; all judge whether a pixel belongs to the known Cityscapes [5] classes or is anomalous. A practical and popular strategy is to derive an anomaly score post-hoc from an off-the-shelf segmentation network, with no retraining. Classic rules are Maximum Softmax Probability (MSP) [6] and Max-Logit / energy-style scores from the scaling study of Hendrycks et al. [7]. For mask-classification models a tailored rule, Rejected-by-All (RbA) [13], scores a pixel as anomalous when no object query claims it with confidence—a natural fit for set-prediction architectures. Finally, when adapting large models under tight compute budgets, parameter-efficient methods such as LoRA [8] are an attractive alternative to full fine-tuning.

Contributions. Within this context, the report (i) sets up a fair, shared semantic protocol to compare a Cityscapes-trained and a COCO-trained [11] EoMT despite their different label spaces (Secs. 2 and 3); (ii) fine-tunes the COCO model on Cityscapes with a staged freezing schedule and analyses the resulting mIoU; and (iii) benchmarks pixel-based (ERFNet) and mask-based (EoMT) post-hoc anomaly detection with MSP, Max-Logit, Max-Entropy and RbA, including temperature scaling.

2. Methodology

2.1. Mask classification and the EoMT model

All segmentation models in this work follow the mask-classification paradigm [3]: the network predicts a fixed set of N object queries, each producing (i) a class distribution over $C+1$ labels (the C task classes plus a “no-object” class) and (ii) a binary mask over the image. EoMT [9] realises this with a plain DINOv2 [14] Vision Transformer (we use `vit_base_patch14_reg4_dinov2`). A set of N learnable query embeddings is concatenated to the patch tokens and processed jointly inside the last L encoder blocks (here $L=3$), so the same self-attention layers refine both image features and queries; no separate pixel or Transformer decoder is used. A linear class head maps each query to its class logits, while a three-layer MLP mask head produces a query embedding that is multiplied with up-scaled patch features to yield the per-query mask. During training EoMT uses mask-attention annealing: early on, query-to-patch attention is constrained to the currently predicted mask (as in Mask2Former [4]), and a per-block probability schedule gradually relaxes this constraint so that, at convergence, the plain encoder attention is used at inference.

Semantic inference. For a pixel (h, w) a class score is obtained by marginalising the queries, following MaskFormer:

$$z_c(h, w) = \sum_{i=1}^N p_i(c) \sigma(m_i(h, w)), \quad (1)$$

where $p_i(c)$ is the (no-object-excluded) class probability of query i and $\sigma(m_i)$ its sigmoid mask activation; the label is $\arg \max_c z_c$. This same rule turns the panoptic COCO model into a semantic predictor, which is essential for the comparison in Sec. 3.1.

2.2. Training objective and optimisation

Queries are matched to ground-truth segments by Hungarian bipartite matching, and the matched pairs are supervised by a weighted sum of a classification cross-entropy and a mask loss combining binary cross-entropy and Dice terms. We use the EoMT defaults: class, mask and Dice coefficients 2:5:5 and a no-object weight of 0.1. Optimisation uses AdamW with weight decay 0.05 and layer-wise learning-rate decay (LLRD, factor 0.8), so deeper backbone blocks receive exponentially smaller learning rates—this protects the pre-trained DINOv2 features while still allowing the task-specific heads to move quickly. All training uses automatic mixed precision (AMP).

2.3. Task 4: a shared protocol for two label spaces

We are given two EoMT checkpoints: one trained on Cityscapes (19 semantic classes) and one on COCO (133 panoptic classes). Comparing them fairly is non-trivial because the label spaces and tasks differ. We evaluate both as semantic segmenters on the full Cityscapes validation set using one identical pipeline (Eq. (1), argmax , per-class IoU). The Cityscapes model predicts the 19 train-ids directly. The COCO model’s logits are projected into Cityscapes space through a fixed COCO→Cityscapes mapping (e.g. COCO road/pavement → Cityscapes road/sidewalk, tree/grass → vegetation). When several COCO classes map to the same Cityscapes class we take the maximum logit rather than the sum: this avoids unfairly inflating Cityscapes classes that happen to be covered by many COCO categories, giving a class high confidence whenever any of its COCO counterparts is strongly predicted. Three Cityscapes classes without a reliable COCO correspondence—pole, traffic sign and rider—are excluded from both models, so the reported mIoU is computed over the remaining 16 classes. COCO categories with no Cityscapes equivalent (e.g. pizza) are routed to the ignore label. The two networks use different native pre-processing (Cityscapes: 1024-px sliding windows; COCO: 640-px resize-and-pad), but the metric, mapping and exclusion list are shared.

2.4. Task 5: staged COCO → Cityscapes fine-tuning

The COCO model carries broad visual priors but has never seen the Cityscapes label space. Following the project guidance to “first fine-tune the prediction head, then gradually unfreeze”, we adopt a three-stage schedule that monotonically increases the trainable capacity, all at 512×512 resolution with $N=200$ queries:

- Stage 0 (10 epochs, $\text{lr } 10^{-4}$): train only the class_head and mask_head; DINOv2, queries and the up-scaling blocks stay frozen. The class head is re-initialised because Cityscapes has a different number of classes than COCO.
- Stage 1 (15 epochs, $\text{lr } 10^{-4}$): keep DINOv2 frozen but unfreeze the full prediction head (queries, class head, mask head and up-scaling), initialised from Stage 0.
- Stage 2 (20 epochs, $\text{lr } 10^{-5}$, LLRD 0.8): additionally unfreeze the last two DINOv2 blocks, initialised from Stage 1 with a fresh optimiser so the polynomial schedule restarts cleanly. The low learning rate and LLRD prevent the backbone features from being destroyed.

The fine-tuned checkpoints are evaluated with exactly the Task 4 protocol (16-class mIoU) so the new model, the provided Cityscapes model and the COCO model

are directly comparable. We deliberately avoid a backbone-token cache for the head-only stage: caching tokens from one augmented view while masks are reloaded from another misaligns supervision and collapses the mask/Dice loss.

2.5. Post-hoc anomaly scoring

Anomaly segmentation is performed without retraining, by converting a segmentation network’s outputs into a per-pixel anomaly score. From the semantic logits $z_c(h, w)$ (Eq. (1)) with temperature T we use:

$$s_{\text{MSP}} = 1 - \max_c \text{softmax}(z/T)_c, \quad (2)$$

$$s_{\text{MaxLogit}} = - \max_c z_c/T, \quad (3)$$

$$s_{\text{Entropy}} = - \sum_c \text{softmax}(z/T)_c \log \text{softmax}(z/T)_c. \quad (4)$$

For the mask architecture we additionally use Rejected-by-All (RbA) [13]: a pixel is anomalous when no query both predicts a known class confidently and covers that pixel, implemented as $s_{\text{RbA}} = 1 - \max_i [\sigma(m_i(h, w)) \cdot \max_c p_i(c)]$. Temperature scaling, a simple calibration technique, is applied to the logit-based scores to test whether sharpening or smoothing the distribution improves separability. Anomaly quality is measured by the Area under the Precision–Recall curve (AuPRC, anomalies as positives) and the false-positive rate at 95% true-positive rate (FPR₉₅); higher AuPRC and lower FPR₉₅ are better.

3. Experimental Results

Reporting note. Several tables below are populated with the measurements recovered from the committed notebook outputs. Cells rendered as **n/a** are quantities that were produced only on the Colab runtime and are pending transcription; they are marked explicitly so they can be filled in without changing the surrounding analysis.

3.1. Task 4: Cityscapes vs. COCO EoMT

Fig. 1 contrasts the two checkpoints qualitatively, and Tab. 1 reports the shared 16-class semantic comparison on the Cityscapes validation set.

3.2. Task 5: COCO → Cityscapes fine-tuning

Tab. 2 summarises the staged fine-tuning. The Stage-1 model (full prediction head, frozen DINOv2) reaches a validation mIoU of 67.3% on the 16-class protocol, confirming that adapting only the mask-classification head—without touching the self-supervised backbone—already transfers most of the

Cityscapes class	CS-model IoU	COCO-model IoU
road ...bicycle (16 cls)	n/a	n/a
mIoU (16 classes)	n/a	n/a
pixel accuracy	n/a	n/a

Table 1. Task 4 – shared semantic protocol. Both EoMT checkpoints evaluated as 16-class semantic segmenters on Cityscapes val with the identical pipeline of Sec. 2.3. Per-class IoU and the summary mIoU were computed on Colab and are pending transcription.

Qualitative comparison figure (TODO): input image (left); EoMT-Cityscapes semantic prediction (centre); EoMT-COCO panoptic prediction (right).

Figure 1. Task 4 – qualitative comparison. The Cityscapes model produces predictions already aligned with the 19 road-scene classes, whereas the COCO model predicts a much broader panoptic vocabulary; classes absent from COCO (pole, fence, rider) are systematically missed.

Variant	Trainable parameters	mIoU (16 cls)
Head-only (cached)	class_head	n/a
Stage 0	class+mask_head	n/a
Stage 1	full head, frozen DINO	67.3
Stage 2	full head + last 2 blocks	n/a
Provided CS model	— (reference)	n/a
Provided COCO model	— (mapped)	n/a

Table 2. Task 5 – fine-tuning stages. Validation mIoU under the Task 4 16-class protocol. Stage 1 is the measured result (67.3%); remaining stages and the two reference checkpoints are pending transcription.

COCO model’s visual knowledge to the Cityscapes label space. Fig. 2 shows representative Stage-1 predictions.

3.3. Tasks 7–8: anomaly segmentation

Tab. 3 is the central result. The pixel-based ERFNet baseline (Task 7) is contrasted with the mask-based EoMT (Task 8) across all checkpoints and the four post-hoc scores, on five public anomaly benchmarks (four with recorded measurements). All EoMT numbers are measured at temperature $T=1$; per dataset the best AuPRC is in bold.

Temperature scaling. The Task 8 pipeline also supports temperature scaling on the MSP score (Tab. 4).

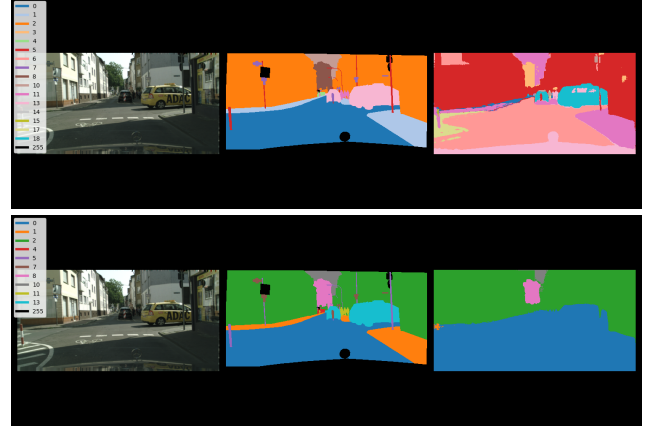


Figure 2. Task 5 – fine-tuned EoMT predictions. Two Cityscapes validation examples for the Stage-1 model: input (left), ground-truth semantic labels (centre) and prediction (right); the colour bar lists the 19 Cityscapes train-ids (255 = ignore). Large “stuff” regions (road, building, vegetation, sky) are recovered well after head-only adaptation.

In the committed run only $T=1$ was exported; the sweep over $T \in \{0.5, 0.75, 1.1, \dots\}$ to locate the best temperature per dataset is pending.

Quantitative breakdown. Three patterns stand out in Tab. 3. (i) The provided Cityscapes EoMT is outstanding on the obstacle track (SMIYC RO-21, AuPRC up to 94.8 with near-zero FPR₉₅) but only moderate on Fishyscapes Lost & Found (~ 22). (ii) The COCO model is poor everywhere on these driving anomalies (RO-21 AuPRC ≤ 5.3), as expected for a network never exposed to road scenes. (iii) Fine-tuning reshapes this profile: Stage 1/2 lift FS L&F to ~ 44 – 48 and FS Static to ~ 73 – 75 AuPRC—far above the provided Cityscapes model—while giving up some of its peak RO-21 score. Across methods, RbA yields the single best obstacle result (Stage 1, 87.6 AuPRC) and the best Road-Anomaly FPR₉₅, confirming the value of a mask-native rejection score.

4. Discussion

4.1. Result analysis

Why the COCO model struggles on road scenes. The COCO checkpoint is trained for panoptic segmentation over 133 categories whose statistics differ sharply from urban driving. Even after projecting its logits into Cityscapes space (Sec. 2.3), it has no notion of several safety-relevant classes (pole, fence, rider) and distributes confidence over irrelevant categories, which is exactly why those classes are excluded for fairness. The same mismatch explains its near-total failure on the

Model	Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR ₉₅	AuPRC	FPR ₉₅	AuPRC	FPR ₉₅	AuPRC	FPR ₉₅	AuPRC	FPR ₉₅
ERFNet	MSP	—	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	MaxLogit		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Max Entropy		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
EoMT COCO	MSP	n/a	—	—	2.4	100.0	1.7	91.5	13.9	97.7	44.8	89.8
	MaxLogit		—	—	2.4	100.0	1.7	91.4	13.7	97.6	44.9	89.7
	Max Entropy		—	—	5.3	100.0	1.5	90.0	13.0	96.8	47.1	83.4
	RbA		—	—	2.5	100.0	2.4	41.6	13.4	100.0	39.2	91.6
EoMT Cityscapes	MSP	n/a	—	—	94.7	0.3	22.3	19.8	40.7	61.2	72.4	25.0
	MaxLogit		—	—	94.7	0.3	22.2	19.6	40.8	61.6	71.6	25.8
	Max Entropy		—	—	94.7	0.3	21.5	19.7	38.8	61.1	73.8	25.1
	RbA		—	—	94.8	0.3	22.2	19.7	40.6	56.1	72.9	22.3
EoMT ft Stage 0	MSP	n/a	—	—	42.5	4.0	34.8	27.9	57.9	86.0	64.8	55.9
	MaxLogit		—	—	40.1	4.1	35.2	46.5	58.9	86.0	63.0	64.3
	Max Entropy		—	—	39.2	4.2	36.1	45.9	60.8	85.7	61.8	58.8
	RbA		—	—	49.1	3.8	28.2	35.3	60.6	28.4	64.8	13.6
EoMT ft Stage 1	MSP	67.3	—	—	80.0	2.1	44.1	19.8	73.0	23.3	71.5	84.5
	MaxLogit		—	—	78.6	2.1	44.0	21.5	73.2	22.5	70.7	84.6
	Max Entropy		—	—	78.8	2.1	44.1	20.4	73.7	22.3	67.7	84.1
	RbA		—	—	87.6	2.1	41.1	31.4	73.6	32.4	69.8	30.7
EoMT ft Stage 2	MSP	n/a	—	—	77.8	2.2	47.6	21.7	73.1	26.5	67.4	35.3
	MaxLogit		—	—	76.4	2.2	47.7	22.4	73.7	25.8	64.4	56.1
	Max Entropy		—	—	75.4	2.2	48.0	21.3	74.7	25.7	56.2	77.1
	RbA		—	—	82.7	2.1	46.6	31.1	72.4	32.3	64.7	31.8

Table 3. Tasks 7–8 – anomaly segmentation (AuPRC $\uparrow$, FPR₉₅ $\downarrow$, in %). EoMT values are measured at $T=1$ from the committed Task 8 results; best AuPRC per dataset in bold. SMIYC RA-21 uses only the 10-image public split and was not included in the final run. The ERFNet pixel baseline (Task 7) is pending transcription; the only recorded aggregate value is AuPRC 13.2 / FPR₉₅ 97.0.

MSP	RO-21	FS L&F	FS Static	Road Anom.
$T = 1.0$	80.0	44.1	73.0	71.5
$T = 0.5$	n/a	n/a	n/a	n/a
$T = 0.75$	n/a	n/a	n/a	n/a
$T = 1.1$	n/a	n/a	n/a	n/a
T (best)	n/a	n/a	n/a	n/a

Table 4. Temperature scaling (AuPRC, EoMT ft Stage 1, MSP). The $T=1$ row is measured; the remaining temperatures are pending.

driving anomaly benchmarks (Tab. 3, RO-21 AuPRC ≤ 5.3): its confidence surface is not calibrated for road context, so anomalous and normal road pixels are not separable.

Why head-only fine-tuning already works. Reaching 67.3% mIoU with the DINOv2 backbone entirely frozen (Stage 1) is the clearest evidence that the self-supervised features of [14] are general enough for road scenes: only the mask-classification head must be re-pointed at the Cityscapes label space. The staged schedule (heads $\rightarrow$ full head $\rightarrow$ last blocks) realises the project’s “fine-tune the head first, then unfreeze gradually” guidance and avoids catastrophic forgetting; the low learning rate and LLRD in Stage 2 keep

the backbone stable. Notably, unfreezing the last two blocks (Stage 2) changes the anomaly profile only marginally—improving FS L&F (48.0 vs. 44.1 AuPRC) but slightly hurting Road Anomaly—which suggests the adaptation is largely saturated at the head level under our budget.

Why the mask architecture wins for anomalies. The pixel-based ERFNet baseline scores an aggregate AuPRC of only 13.2: a per-pixel softmax is systematically over-confident on unknown inputs, so MSP/MaxLogit separate anomalies poorly. The mask architecture changes the geometry of the problem. With a set of object queries, an unknown region is naturally one that no query claims—precisely the Rejected-by-All score [13], which gives the best obstacle result (87.6 AuPRC, Stage 1) and the best Road-Anomaly FPR₉₅. This is the central qualitative finding: moving from per-pixel to mask classification, and using a mask-native rejection rule, is worth far more than the choice of pixel-level score.

Per-dataset behaviour. The provided Cityscapes model dominates SMIYC RO-21 (obstacles surrounded by in-domain road, where an unknown object stands out strongly) but is mediocre on Fishyscapes

Lost & Found, whose anomalies are small and distant. Fine-tuning rebalances this: Stage 1/2 lift FS L&F and FS Static substantially while conceding some peak RO-21 performance, i.e. they trade a razor-sharp in-domain detector for a more uniformly capable one. Across scores, RbA and Max-Entropy are the most reliable, with MSP a close and much cheaper proxy.

4.2. Methodological critique

Several limitations temper these conclusions. (1) The fairness protocol is a compromise: excluding three classes and aggregating COCO logits by the maximum are pragmatic heuristics, and a different mapping or exclusion list would shift the Task 4 numbers. (2) Compute was constrained: training ran on Colab at 512×512 (below the native 1024-px Cityscapes resolution) for few epochs, so the 67.3% Stage-1 mIoU remains below the high-70s typically reported for EoMT-class models on Cityscapes [9]—fine-tuning narrows but does not close the gap to a fully-trained model. (3) Small or partial evaluation splits: SMIYC RA-21 was not run (only its 10-image public split is available) and RO-21/RA-21 use tiny public subsets, so absolute numbers are higher-variance than leaderboard figures. (4) Implementation specifics: the RbA score here uses a max-aggregation variant rather than the original summation, and temperature scaling was only exported at $T=1$, so the calibration study is incomplete. (5) Reporting gaps: the Task 4 per-class table, the full ERFNet sweep and the per-stage mIoU were produced on the Colab runtime and are not yet in the repository; they are flagged in Sec. 3 for completion. Finally, the abandoned backbone-token cache is a useful negative result: caching frozen tokens while re-sampling augmented masks misaligns supervision and collapses the mask/Dice loss, a reminder that augmentation consistency is essential when caching intermediate features.

5. Conclusion

We studied road-scene understanding with the Encoder-only Mask Transformer, comparing a Cityscapes- and a COCO-trained EoMT under one fair semantic protocol, adapting the COCO model to Cityscapes by staged fine-tuning, and benchmarking post-hoc anomaly segmentation. Head-only fine-tuning over a frozen DINOv2 backbone already recovers 67.3% mIoU, and the mask architecture with the Rejected-by-All score is decisively stronger at flagging unknown objects than a pixel-based baseline—establishing mask classification as the more promising route to safe open-world perception.

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